

Research on state evaluation and risk assessment for relay protection system based on machine learning algorithm

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Abstract: The relay protection system plays an important role in ensuring the stable operation of power systems. Combined with operation data collected from a region in China, this study is aimed at providing a reliable quantitative basis for relay protection systems' operating maintenance by the aid of a semi-supervised Mahalanobis distance machine learning algorithm. The evaluation result is first applied as a training set on the basis of the analytic hierarchy process fuzzy synthetic evaluation. Then, contrastive analysis is conducted in terms of accuracy, processing time, and feasibility. It includes comparative cases with a supervised multiple regression analysis algorithm and unsupervised K-means algorithm. The comparison result reveals that the algorithm can effectively and accurately predict the running state of the equipment and offer a quantitative reference for relative maintenance strategy.

1 Introduction

Over the past two decades, there has become an increasing number of smart power electronic devices and new energy equipment accessed into power systems [1, 2]. It is of significance for stable electric power systems operation to improve the maintenance efficiency of power system equipment [3, 4].

Correct, fast, and reliable action of protective relays will avoid power systems from aggravation when failures occur, thus efficaciously enhancing the stability of power systems and reducing economic loss. Protective relaying equipment provides effective security for power systems operation [5]. Si *et al.* [6] constructed a failure probability model of relay protection system due to a hidden fault in power networks with electromagnetic loops. The ORNL-PSerc-Alaska (OPA) model is a blackout model proposed by researchers at Oak Ridge National Laboratory. Taking value at risk (VaR) and conditional VaR (CVaR), the influence on relay protection VaR and CVaR caused by blackout is analysed by applying the OPA model in [7]. Liu *et al.* [8] studied the effectiveness of the proposed risk assessment method using a modified 9-bus system and the IEEE 69-bus system which is applicable to extreme cases in power systems. Jedrzejczak and Anders [9] addressed an issue of reliability evaluation of protective relays, operating in a harmonic polluted environment. It presents a solution to the problem of finding the transition rate matrix for a Markov chain of a simplified digital distance relay model. Jedrzejczak *et al.* [10] developed a comprehensive Markov reliability model to categorise the possible functional states of a component protected by a relay operating in a harmonic-polluted environment.

Machine learning technology is widely used in the power system. For instance, Dubey *et al.* [11] presented an extreme learning machine (ELM) based on a fast and accurate adaptive relaying scheme for stand-alone distance protection of the transmission network. The proposed ELM based on a fast adaptive distance relaying scheme and adaptive distance relaying scheme is extensively validated on two terminal transmission lines with complex mutual coupling and shunt capacitance. Machine learning methods, such as artificial neural networks, K-nearest neighbours, and support vector machine (SVM) were employed to classify lightning transient signals in [12, 13], considering mother wavelet and different decomposition levels of three-phase voltages. A multi-objective dragonfly algorithm was developed in [14] to optimise the Elman neural network model for the electrical power

system forecasting to overcome the drawbacks of the single-objective optimisation algorithm. Li *et al.* [15] proposed an approach based on the combined wavelet transform-ELM technique for fault section identification to tell whether the fault is before or after the series capacitor in a series compensated transmission line. A hybrid model based on synchronous optimisation for multi-step short-term wind speed forecasting was established by Mishra *et al.* [16]. Dubey *et al.* [17] demonstrated an ELM based on a fast and accurate adaptive distance relaying scheme for transmission lines in the presence of a static synchronous series compensator. Kumar *et al.* [18] built the fault prediction model using the least squares SVM learning method associated with linear, polynomial, and radial basis function kernel functions applying ten different feature selection techniques to choose the best set of metrics from a set of 20 metrics. Three models of machine learning including a multilayer feed-forward neural network, support vector regression with a radial basis function, and adaptive neuro-fuzzy inference system are implemented to predict wind speed, wind direction, and output power of a wind turbine in [19, 20]. Sadegh and Farahani [21] provided a learning machine algorithm based on Lyapunov stability theory for neural networks using a two-area four-machine power system and IEEE 39-bus power system. Zhang *et al.* [22] proposed a method for data-based line trip fault prediction in power systems using long short-term memory networks and SVM. However, existing methods have several non-negligible limitations: (i) the advanced algorithms are required to improve pattern recognition accuracy; (ii) the evaluation methods have poor self-adaptability and self-correction; (iii) highly subjective for schemes and without much data support; (iv) simulation results from some models may not be able to help much for practical operation and maintenance. To address the aforementioned gaps, in this study, we develop a risk assessment model for relay protection systems based on a semi-supervised Mahalanobis distance (MD) machine learning algorithm. The main innovations and contributions of this study include:

- (i) Characterising and quantifying the state evaluation model for the relay protection system to provide training sets considering both dynamic and static indices.
- (ii) Developing a MD machine learning algorithm to attain the risk assessment result of relay protection systems, which is more suitable for actual conditions.

- (iii) Comparing and analysing the diagnosis results of supervised, unsupervised, and semi-supervised machine learning methods.
- (iv) Offering a corresponding maintenance strategy on the basis of evaluation results.
- (v) The remainder of the paper is organised as follows. Section 2 develops the status evaluation model for the relay protection system. A machine learning algorithm for the operation state of the relay protection system is described in Section 3. A maintenance strategy is presented in Section 4. Conclusion and future work are given in Section 5.

2 Status evaluation of relay protection system

In conjunction with running records of relay protection equipment in a certain region of China Southern Power Grid Corporation, the operation state evaluation model for relay protection systems is put forward based on the analytic hierarchy process (AHP). Computing results can be provided as training sets for the machine learning algorithm, thus help to develop a maintenance strategy on the foundation of quantitative calculation results.

2.1 Evaluation indicator

In accordance with DL/T860 Engineering implement standard. Network topology of relay protection system intelligent substation is displayed in Fig. 1. The equipment is divided into three levels, i.e. station level, bay level, and process level [23–25].

- *Station level:* Station level contains a background monitor, telecontrol machine, clock system, communication power supply system etc. It mainly achieves the monitoring functions for running conditions, alarm information, CPU, memory utilisation, hard disk usage, and network interface.
- *Bay level:* Equipment in bay level includes a protective device, measure-control (M-C) device, fault oscillography, network analyser etc. Its main function is monitoring A/D, generic object-oriented substation event (GOOSE), sample value (SV), and I/O.
- *Process level:* Process level consists of merging unit, intelligent terminal etc. Its primary task is to collect information about clock matching, chip condition, and luminous power.

Through widely surveyed and studied from manufactures and R&D personnel, static and real-time parameters reflecting running state of relay protection systems were selected after referring to guidance on state evaluation and risk assessment of secondary equipment of China southern power grid corporation and Intelligent substation relay protection information standard. Taking Table 1 as an example, the assessment indices of M-C devices are displayed as follows:

M-C device is significant for dispatchers to grasp information of substation and control it. Its assessment indicator contains 28 indices, which is shown in Table 1. Merging unit, with 18 evaluation indicators, can realise the collection and transmission of

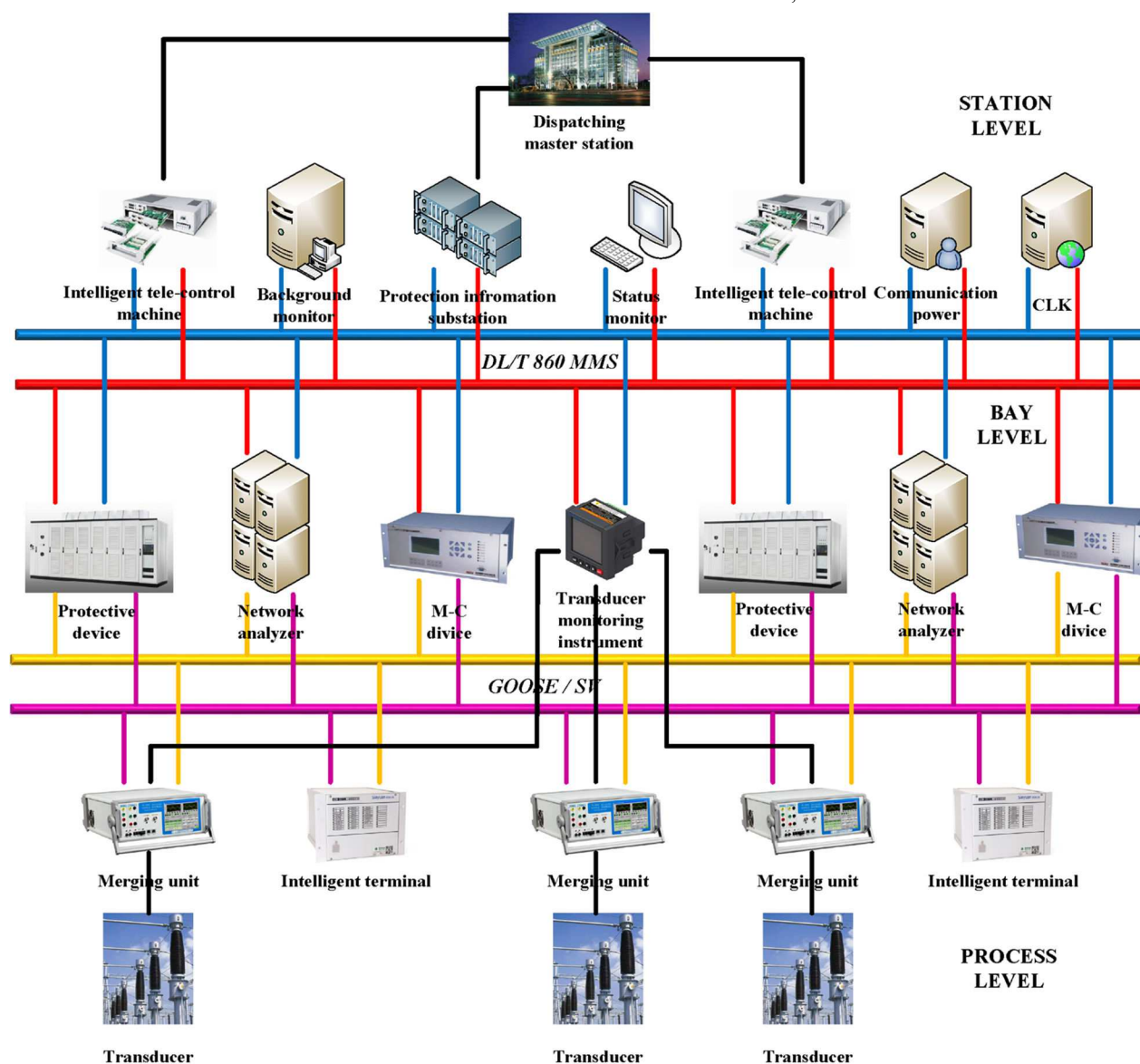


Fig. 1 Network topology of the relay protection system

Table 1 Evaluation indicator of M-C device

Sequence number	Indicator	Importance degree	Data type	Indicator type
1	ROM mistake	4	D	cost
2	RAM mistake	4	D	cost
3	output mistake	3	D	cost
4	input mistake	3	D	cost
5	setting value mistake	3	D	cost
6	relaying plate mistake	4	D	cost
7	Electrically Erasable Programmable Read-Only Memory (EEPROM) mistake	3	D	cost
8	communication outage of serial port	3	D	cost
9	system misconfiguration	3	D	cost
10	communication outage of CPU	4	D	cost
11	configuration table mistake	3	D	cost
12	logic table mistake	3	D	cost
13	Tele-indication deflection inconformity	2	D	cost
14	flash mistake	2	D	cost
15	system operation abnormal	3	D	cost
16	circuit breaker's non-fault-trip	3	D	cost
17	communication outage of bay level	3	D	cost
18	optical power abnormal	2	D	cost
19	internal crystal oscillator frequency deviation	3	A	interval
20	alarm information of GOOSE (unusual)	3	D	cost
21	alarm information of GOOSE (important)	4	D	cost
22	alarm information of GOOSE (error)	4	D	cost
23	alarm information of SV (unusual)	3	D	cost
24	alarm information of SV (important)	4	D	cost
25	alarm information of SV (error)	4	D	cost
26	alarm information of MMS (unusual)	3	D	cost
27	alarm information of MMS (important)	4	D	cost
28	alarm information of MMS (error)	4	D	cost

Table 2 Nine-level judgment table

Implication	Degree	Implication	Degree	Implication	Degree
paramount	9	important	3	not quite important	1/5
very important	7	equal	1	not very important	1/7
quite important	5	unimportant	1/3	the least important	1/9

voltage and current data in terms of time series. The function of high-, medium-, and low-voltage line all-in-one protection machine is to remove fault equipment and prevent expanding fault. The intelligent terminal is in charge of circuit breakers instructions and equipment positioning.

2.2 Index weight value

2.2.1 Evaluation indicators of the relay protection system: Evaluation indicators for relay protection system operation consist of dynamic evaluation indicators and static evaluation indicators. The dynamic indicator of intelligent relay protection equipment derives from self-check information mainly including memory, setting value, signal input and output, relaying plate, communication interface, CPU, and so on. The static indicator derives from historical operation record mainly including alarm information record of GOOSE web, SV web, and manufacturing message specification (MMS) web. Alarm information is classified as normal, warning, error, and critical state.

2.2.2 Judgment matrix: Nine-level judgment table [26] is used to calculate the relevant important degree.

As shown in Table 2, degree '1' means x_i and x_j are equally important; '3' represents that x_i is slightly more important than x_j ; '5' signifies that x_i is obviously more important than x_j ; '7' denotes

that x_i is significantly more important than x_j ; '9' implies that x_i is extremely more important than x_j [27].

To get the influence degree of indicator X containing m indices on Y , the importance degree comparison matrix is established after pairwise comparison. Define the comparison coefficient as $a_{ij} = x_i/x_j$, judgment matrix A can be obtained after C_n^2 times comparison process

$$A = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1m} \\ a_{21} & 1 & \cdots & a_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & 1 \end{bmatrix}_{m \times m} \quad (1)$$

Maximum eigenvalue and corresponding eigenvector of the judgment matrix are attained from formulas (2)–(6)

$$M_i = \prod_{j=1}^n a_{ij} \quad i = 1, 2, \dots, n \quad (2)$$

n th root of M_i is written as

$$\bar{w}_i = M_i^{\frac{1}{n}} \quad i = 1, 2, \dots, n \quad (3)$$

Normalisation for $\bar{w}_i$

Table 3 Satty random index table

N	RI	N	RI	N	RI	N	RI
1	0	8	1.41	15	1.59	22	1.64
2	0	9	1.46	16	1.60	23	1.65
3	0.58	10	1.49	17	1.61	24	1.65
4	0.90	11	1.52	18	1.61	25	1.66
5	1.12	12	1.54	19	1.62	26	1.66
6	1.24	13	1.56	20	1.63	27	1.66
7	1.32	14	1.58	21	1.64	28	1.67

$$\omega'_i = \bar{w}_i / \left(\sum_{j=1}^n \bar{w}_j \right) \quad i = 1, 2, \dots, n \quad (4)$$

The maximum eigenvalue λ can be calculated as:

$$\lambda = \frac{1}{n} \sum_{i=1}^n \left[\frac{\sum_{j=1}^n a_{ij} w_j}{w_i} \right] \quad (5)$$

Then the eigenvector is expressed as

$$W = [\omega'_1, \omega'_2, \dots, \omega'_n] \quad (6)$$

2.2.3 Weight calculation and consistency check: The weight calculation result is considered to be correct only if the judgement matrix satisfies consistency check criteria. Consistency check criteria are formulated as

$$CI = (\lambda - N) / (N - 1) \quad (7)$$

where λ is the maximum eigenvalue of n th-order positive reciprocal matrix A ; N is the non-zero characteristic root of the n th-order consistent matrix.

Consistency ratio (CR) is defined as

$$CR = CI / RI \quad (8)$$

where RI is a random index.

Satty random index is shown in Table 3.

If $CR < 0.10$, the consistency check of the judgment matrix is considered to be accepted. CR calculation result of disparate equipment is as $CR_{m-c} = 0.0069$, $CR_m = 0.0123$, $CR_p = 0.0118$, $CR_i = 0.0122$. Consistency check criteria are satisfied.

Judgment matrix of a different kind of device can be deduced through transformation

$$A_{m-c} = \begin{bmatrix} 1 & 1 & 4 & 4 & \dots & 1 \\ 1 & 1 & 4 & 4 & \dots & 1 \\ 0.25 & 0.25 & 1 & 1 & \dots & 0.25 \\ 0.25 & 0.25 & 1 & 1 & \dots & 0.25 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 1 & 4 & 4 & \dots & 1 \end{bmatrix}_{28 \times 28} \quad (9)$$

$$A_m = \begin{bmatrix} 1 & 4 & 6 & 6 & \dots & 1 \\ 0.25 & 1 & 3 & 3 & \dots & 0.25 \\ 0.17 & 0.33 & 1 & 1 & \dots & 0.17 \\ 0.17 & 0.33 & 1 & 1 & \dots & 0.17 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 4 & 6 & 6 & \dots & 1 \end{bmatrix}_{18 \times 18} \quad (10)$$

$$A_p = \begin{bmatrix} 1 & 1 & 1 & 4 & \dots & 1 \\ 1 & 1 & 1 & 4 & \dots & 1 \\ 1 & 1 & 1 & 4 & \dots & 1 \\ 0.25 & 0.25 & 0.25 & 1 & \dots & 0.25 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 1 & 1 & 4 & \dots & 1 \end{bmatrix}_{24 \times 24} \quad (11)$$

Table 4 Index weight calculations of different device

Sequence number of indicator	M-C device	Merging unit	Protective device	Intelligent terminal
1	0.0963	0.1283	0.1049	0.1311
2	0.0963	0.0160	0.1049	0.0101
3	0.0024	0.0036	0.1049	0.0013
4	0.0024	0.0036	0.0057	0.0013
5	0.0024	0.0036	0.0057	0.0013
6	0.0963	0.0036	0.0057	0.0013
7	0.0024	0.0036	0.0008	0.0013
8	0.0024	0.0036	0.0008	0.0013
9	0.0024	0.0160	0.0057	0.0013
10	0.0963	0.0160	0.0057	0.0013
11	0.0024	0.1283	0.0008	0.0101
12	0.0024	0.1283	0.0008	0.0101
13	0.0002	0.0160	0.0008	0.0013
14	0.0002	0.1283	0.0008	0.0101
15	0.0024	0.1283	0.0057	0.0101
16	0.0024	0.0160	0.0057	0.1311
17	0.0024	0.1283	0.1049	0.1311
18	0.0002	0.1283	0.1049	0.0101
19	0.0024	—	0.0057	0.1311
20	0.0024	—	0.1049	0.1311
21	0.0963	—	0.1049	0.0101
22	0.0963	—	0.0057	0.1311
23	0.0024	—	0.1049	0.1311
24	0.0963	—	0.1049	—
25	0.0963	—	—	—
26	0.0024	—	—	—
27	0.0963	—	—	—
28	0.0963	—	—	—

$$A_i = \begin{bmatrix} 1 & 4 & 6 & 6 & \dots & 1 \\ 0.25 & 1 & 3 & 3 & \dots & 0.25 \\ 0.17 & 0.33 & 1 & 1 & \dots & 0.17 \\ 0.17 & 0.33 & 1 & 1 & \dots & 0.17 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & 4 & 6 & 6 & \dots & 1 \end{bmatrix}_{23 \times 23} \quad (12)$$

Index weight calculations of the different device using formulas (2)–(6) are revealed in Table 4.

2.3 Fuzzy synthetic evaluation

Considering there exists a fuzzy relationship between variables, actual operation cannot be described simply by 0–1 logic when it comes to evaluating the running status of relay protection systems. The fuzzy relationship should be represented by the continuous interval $[0, 1]$ instead of the binary logic of 0 (false) and 1 (true).

2.3.1 Fuzzy synthetic evaluation characteristic domain: The characteristic domain of relay protection fuzzy contains index set, weight set, and assessment set. The index set, denoted as $I = \{i_1, i_2,$

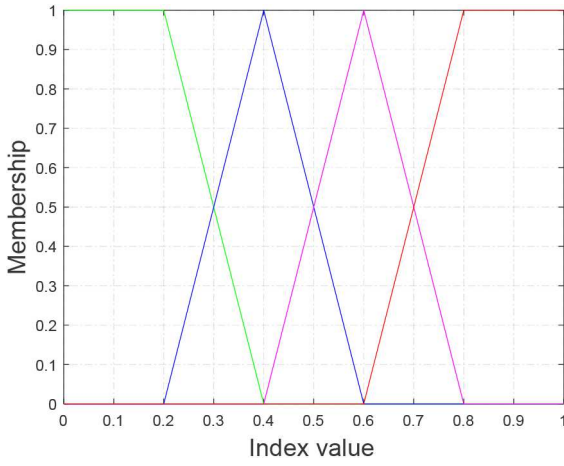


Fig. 2 Triangle distribution membership

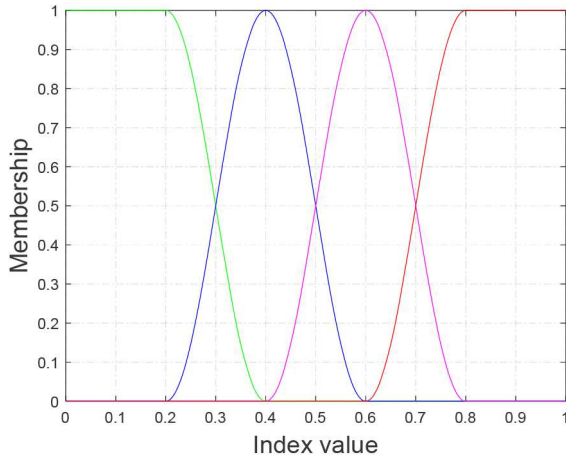


Fig. 3 Ridge distribution membership

..., i_n containing n indicators, refers to evaluation indices used to state assessment of relay protection system. Weight value of the indicators, expressed as $W = \{w_1, w_2, \dots, w_n\}$, can be computed based on AHP. Assuming the assessment set consists of m evaluation results, then it can be written as $E = \{e_1, e_2, \dots, e_m\}$.

2.3.2 Data standardisation processing: To conduct data standardisation processing to convert the actual value to the interval $[0, 1]$, state evaluation indices can be divided into three classes containing benefit type, cost type, and interval type in light of exchange ways.

(i) **Benefit type:** Benefit type means the larger the value is, the better it will be. Converting function of benefit type $g_b(x)$ is

$$g_b(x) = \begin{cases} 0, & x < x_{\min} \\ (x - x_{\min}) / (x_{\max} - x_{\min}), & x_{\min} \leq x \leq x_{\max} \\ 1, & x > x_{\max} \end{cases} \quad (13)$$

where x is the actual operation value, $[x_{\min}, x_{\max}]$ is the lower limit and upper limit of corresponding interval.

(ii) **Cost type:** Cost type means the less, the better. Converting function of cost type $g_c(x)$ is

$$g_c(x) = \begin{cases} 1, & x < x_{\min} \\ (x_{\max} - x) / (x_{\max} - x_{\min}), & x_{\min} \leq x \leq x_{\max} \\ 0, & x > x_{\max} \end{cases} \quad (14)$$

where x is the actual operation value, $[x_{\min}, x_{\max}]$ is the lower limit and upper limit of corresponding interval.

(iii) **Interval type:** Interval type means the value is supposed to be in the ideal interval. Converting function of interval type $g_i(x)$ is

$$g_i(x) = \begin{cases} 0, & x < x_{\min} \cup x > x_{\max} \\ (x - x_{\min}) / (x_{i_{\max}} - x_{\min}), & x_{\min} \leq x \leq x_{i_{\max}} \\ 1, & x_{i_{\min}} \leq x \leq x_{i_{\max}} \\ (x_{\max} - x) / (x_{\max} - x_{i_{\min}}), & x_{i_{\max}} \leq x \leq x_{\max} \end{cases} \quad (15)$$

where x is the actual operation value, $[x_{i_{\min}}, x_{i_{\max}}]$ is the ideal operation interval, $[x_{\min}, x_{\max}]$ is the required operation interval.

2.3.3 Fuzzy synthetic membership function: Triangle distribution and ridge distribution membership functions are widely used in engineering practice. The operation state of the relay protection system is divided into four levels, i.e. normal, warning, error, and critical state. Triangle distribution membership function of the different state is as follows:

$$h_{t-1}(u) = \begin{cases} 1, & u \leq 0.2 \\ -(u - 0.4) / 0.2, & 0.2 < u \leq 0.4 \\ 0, & u > 0.4 \end{cases} \quad (16)$$

$$h_{t-2}(u) = \begin{cases} 0, & u \leq 0.2 \cup u > 0.6 \\ (u - 0.2) / 0.2, & 0.2 < u \leq 0.4 \\ -(u - 0.6) / 0.2, & 0.4 < u \leq 0.6 \end{cases} \quad (17)$$

$$h_{t-3}(u) = \begin{cases} 0, & u \leq 0.4 \cup u > 0.8 \\ (u - 0.4) / 0.2, & 0.4 < u \leq 0.6 \\ -(u - 0.8) / 0.2, & 0.6 < u \leq 0.8 \end{cases} \quad (18)$$

$$h_{t-4}(u) = \begin{cases} 0, & u \leq 0.6 \\ (u - 0.6) / 0.2, & 0.6 < u \leq 0.8 \\ 1, & u > 0.8 \end{cases} \quad (19)$$

Triangle distribution membership function calculation model combined with formulas (16)–(19) is shown in Fig. 2.

Ridge distribution membership function of the different state is as follows:

$$h_{r-1}(u) = \begin{cases} 1, & u \leq 0.2 \\ 0.5 - 0.5\sin\pi(u - 0.3) / 0.2, & 0.2 < u \leq 0.4 \\ 0, & u > 0.4 \end{cases} \quad (20)$$

$$h_{r-2}(u) = \begin{cases} 0, & u \leq 0.2 \cup u > 0.6 \\ 0.5 + 0.5\sin\pi(u - 0.3) / 0.2, & 0.2 < u \leq 0.4 \\ 0.5 - 0.5\sin\pi(u - 0.5) / 0.2, & 0.4 < u \leq 0.6 \end{cases} \quad (21)$$

$$h_{r-3}(u) = \begin{cases} 0, & u \leq 0.4 \cup u > 0.8 \\ 0.5 + 0.5\sin\pi(u - 0.5) / 0.2, & 0.4 < u \leq 0.6 \\ 0.5 - 0.5\sin\pi(u - 0.7) / 0.2, & 0.6 < u \leq 0.8 \end{cases} \quad (22)$$

$$h_{r-4}(u) = \begin{cases} 0, & u \leq 0.6 \\ 0.5 + 0.5\sin\pi(u - 0.7) / 0.2, & 0.6 < u \leq 0.8 \\ 1, & u > 0.8 \end{cases} \quad (23)$$

Ridge distribution membership function calculation model combined with formulas (20)–(23) is displayed in Fig. 3.

It can be seen from Figs. 2 and 3 that compared with the triangular distribution membership function, ridge distribution encircles higher membership information when the index is in a high membership position. Meanwhile, ridge distribution eliminates less membership information when the index is in a low membership position. Triangle distribution fuzzy membership tends to get a conservative result. Allowing for the analysis above, ridge distribution is applied to calculate the membership matrix of each state in this study.

Table 5 Equipment record of protective device

ID	Device name	Device type	Mfrs.	Transformer substation (TS)
1747	110 kV circuit I protection	PCS-943AM_111255	NARI	110 kV Qingshan
1750	110 kV circuit II protection	PCS-943AM_111255	NARI	110 kV Qingshan
1760	110 kV bus protection	PCS-915_111038	NARI	110 kV Qingshan
1762	110 kV automatic bus transfer equipment	PCS-9651DA-D_00118202	NRR	110 kV Qingshan
1763	No. 1 main transformer protection I	PCS-978EA_111257	NARI	110 kV Qingshan
1764	No. 1 main transformer protection II	PCS-978EA_111257	NARI	110 kV Qingshan
1774	No. 2 main transformer protection I	PCS-978EA_111257	NARI	110 kV Qingshan
1775	No. 2 main transformer protection II	PCS-978EA_111257	NARI	110 kV Qingshan
1787	No. 3 main transformer protection I	PCS-978EA_111257	NARI	110 kV Qingshan
1788	No. 3 main transformer protection II	PCS-978EA_111257	NARI	110 kV Qingshan
...	...	...	...	...
1874	No. 1 transformer CSC241C	CSC241C V3.08 15bc	SIFANG	110 kV Ziguang

Table 6 Data standardisation criteria of protection device

Indicator sequence	Weight	$x_{\max}$	$x_{\min}$	$x_{i_{\max}}$	$x_{i_{\min}}$
1	0.1049	1	0	0	0
2	0.1049	1	0	0	0
3	0.1049	1	0	0	0
4	0.0057	1	0	0	0
5	0.0057	1	0	0	0
6	0.0057	1	0	0	0
7	0.0008	1	0	0	0
8	0.0008	1	0	0	0
9	0.0057	1	0	0	0
10	0.0057	1	0	0	0
11	0.0008	1	0	0	0
12	0.0008	1	0	0	0
13	0.0008	1	0	0	0
14	0.0008	1	0	0	0
15	0.0057	50	-50	-30	30
16	0.0057	100	0	0	0
17	0.1049	100	0	0	0
18	0.1049	100	0	0	0
19	0.0057	100	0	0	0
20	0.1049	100	0	0	0
21	0.1049	100	0	0	0
22	0.0057	100	0	0	0
23	0.1049	100	0	0	0
24	0.1049	100	0	0	0

2.3.4 Getting training sets: Data information is extracted from the China Southern Power Grid Corporation including a M-C device, merging unit, protective device, and intelligent terminal from 2014 to 2017. Take 80 installations of 110 kV Qingshan transformer and 110 kV Ziguang transformer in Guizhou province, for instance, equipment record is revealed in Table 5.

As shown in Table 5, equipment ledger can be obtained by an energy management system, outage management system, production management system, and distribution management system.

The number of indicators utilised to evaluate protection device is $n_p = 24$; Assessment grade type $m_p = 4$; index set $I_p = \{\text{equipment parameter mistake, ROM mistake, system setting value mistake, ...}\}$; assessment set is $E_p = \{\text{normal, warning, error, critical}\}$; weight set is $W_p = \{0.1049, 0.1049, 0.1049, 0.0057, \dots, 0.1049\}$.

Data standardisation processing criteria of the protection device are displayed in Table 6.

Indicator databases revealed in Table 6 are obtained from station level, bay level, and process level: (i) operating condition and warning are sensed by tele-controllers. Data interaction between station level is realised by DL/860. (ii) M-C devices and protective devices collect data form the MMS network. Switchers acquire data via DL/T860. (iii) Self-monitoring information of the

intelligent terminal and merging unit gained by equipment chip is sent to the M-C device through GOOSE. Data collected by the M-C device is uploaded to the online status monitoring module of station level through MMS. Network analyser attained and analyses GOOSE and SV messages in the process level.

Taking device 1747 on 1 June 2017 for instance, operating state assessment for 110 kV circuit I protection of Qingshan transformer can be calculated as shown in Fig. 4.

Fig. 4a shows the corresponding membership of each index of protective device 1747. Fig. 4b shows the membership radar graph with the weight taken into account. The dotted line represents the upper limit of the ideal upper limit of the evaluation index. The solid line denotes the actual evaluation result.

Training sets prepared for machine learning algorithms are composed of 100 cycles of 80 protective devices evaluation results. It is listed in Table 7.

s , as the scale of the machine learning training set, represents the proportion of the overall sample. The training set contains the samples with the sequence as $[1, 8000s]$. As shown in Table 7, the samples whose sequence is $[8000s + 1, 8000]$ make up the test set.

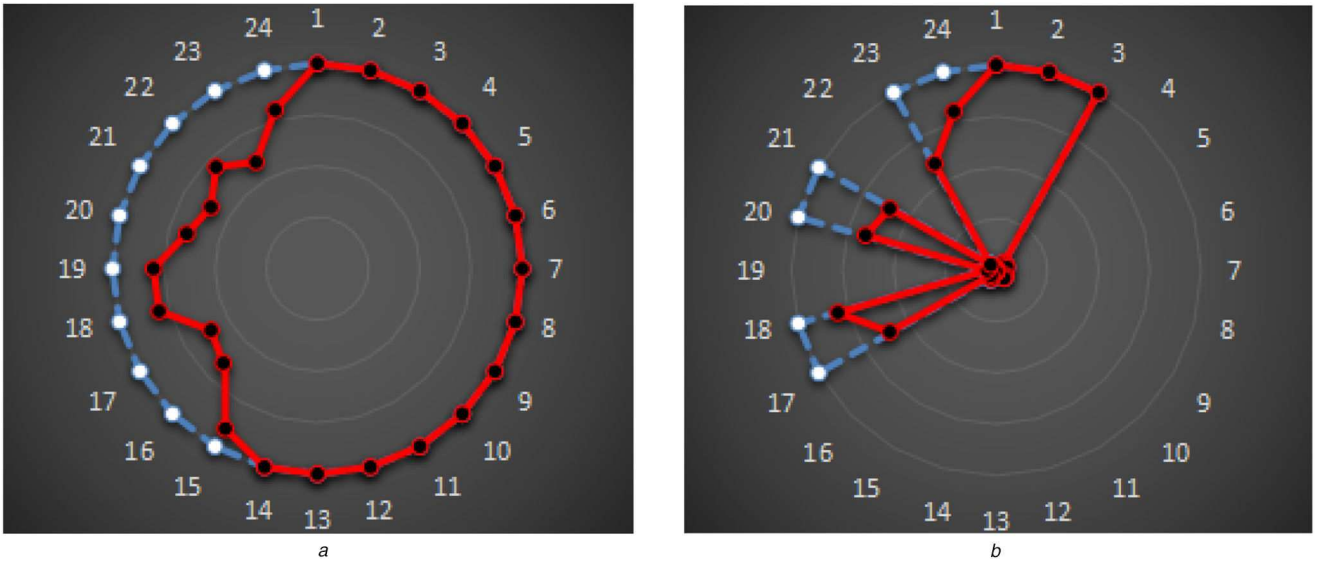


Fig. 4 Operating state assessment diagram for Qingshan 110kV circuit I transformer protection
(a) Indicator radar map of device 1747,
(b) Indicator radar map of device 1747 considering weight

Table 7 Assessment result of protective devices

ID	Evaluation cycle	Index 1	...	Index 24	Score	Grade
1747	2017/06/01	0	...	4	94.99	4
1750	2017/06/13	0	...	10	90.93	4
1768	2017/07/02	0	...	5	88.36	3
...	...	...	...	...	...	...
1810	2017/07/06	0	...	20	94.29	4

3 Machine learning algorithm for operation state of the relay protection system

There are four kinds of devices assessed by using the AHP-fuzzy synthetic evaluation method, and the number of each kind of device is $d_i = \{80, 10, 24, 33\}$. The amount of assessment cycles is c_i , p_i represents the proportion of the overall sample. $d_i c_i p_i$ stands for the number of training sets of separate equipment, and $d_i c_i (1 - p_i)$ refers to the number of a test set of each kind of device. To overcome subjectivity defects when using a fuzzy comprehensive algorithm, a machine learning method is applied in this study to make the calculation result more realistic.

3.1 Supervised multiple regression analysis algorithms

With a view to diverse types of equipment, multiple regression equations for the protective device, M-C device, merging unit, and intelligent terminal can be written as

$$Y_i = f(X_i, B_i, \varepsilon_i) \quad (i = 1, 2, 3, 4) \quad (24)$$

where Y_i is the evaluation result of different devices; X_i is the assessment index; B_i is the regression coefficient; ε_i is the random error.

To minimise the residual sum of squares in the historical evaluation cycle, optimisation objective function is proposed as

$$\min \text{RSS} = \sum_{j=1}^n \varepsilon_j^2 = \sum_{j=1}^n (Y_{1j} - \hat{Y}_{1j})^2 \quad (25)$$

where Y_{1j} is the observed value of protective device, $\hat{Y}_{1j}$ is the estimated value of protective device. In consideration of the extremum principle, constraint function can be assigned as (see (26)), where j is the number of the training set. Taking the statistics of 100 evaluation cycles of 80 protective devices as a training set with training set proportion $p_1 = 80\%$, residual level diagram can be attained by using supervised multiple regression analysis algorithms with training set as shown in Fig. 5.

When significance level $\alpha = 0.05$, there are 308 outlier sample points in the residual leverage graph as shown in Fig. 5 (red lines). The sample pass rate is 95.19%.

After removing the red outliers in the figure, a modified regression residual leverage diagram with data cleaning is displayed in Fig. 6.

The number of outlier samples in the revised regression model is 274. The passing rate is raised to 95.50%.

3.2 Unsupervised k -means algorithm based on Euclidean distance (ED)

Take protective devices, e.g. unsupervised k -means algorithm based on ED is adopted to handle unlabelled data. Then the

$$\text{s.t.} \quad \begin{cases} \frac{\partial \text{RSS}}{\partial b_0} = -2 \sum [Y_1 - (b_0 + b_1 x_{1j} + b_2 x_{2j} + \dots + b_{24} x_{24j})] = 0 \\ \frac{\partial \text{RSS}}{\partial b_1} = -2 x_1 \sum [Y_1 - (b_0 + b_1 x_{1j} + b_2 x_{2j} + \dots + b_{24} x_{24j})] = 0 \\ \dots \\ \frac{\partial \text{RSS}}{\partial b_{24}} = -2 x_{24} \sum [Y_1 - (b_0 + b_1 x_{1j} + b_2 x_{2j} + \dots + b_{24} x_{24j})] = 0 \end{cases} \quad (26)$$

evaluation index is $X_1 = \{x_1, x_2, \dots, x_{24}\}$. The ED between samples a and b in the training set is

$$d(x_a, x_b) = \sum_{j=1}^n (x_{aj} - x_{bj})^2 \quad (27)$$

where $n = 24$. The target function is set up as

$$\min D_C = \sum_{k=1}^K \sum_{l=1}^{n_k} \left\| x_l - m_k \right\|^2 \quad (28)$$

where k is the number of clustering; n_k is the number of the sample; m_k is the mean value of assorted samples.

Flowchart of unsupervised K-means algorithm based on ED is shown in Fig. 7

(i) The number of initial centre clusters is $k = 4$. To select the initial centre point of the protective device training set, two samples x_{i1} and x_{i2} with the longest distance from training sets are picked out to be the first and second initial centre meeting the objective function

$$d(x_{i1}, x_{i2}) = \max_{x_{ij} \in T, x_{il} \in T} [d(x_{ij}, x_{il})] \quad (29)$$

where x_{ij} and x_{il} are any two samples from training sets T . Then select the third initial clustering point x_{i3} , which is satisfied with the condition that

$$\min_{r=1,2} [d(x_{i3}, x_{ir})] = \max_{r=1,2} [\min_{x_{ij} \in T} [d(x_{ij}, x_{ir})]] \quad (30)$$

Similarly, choose the fourth initial clustering point x_{i4} meeting the condition that

$$\min_{r=1,2,3} [d(x_{i4}, x_{ir})] = \max_{r=1,2,3} [\min_{x_{ij} \in T} [d(x_{ij}, x_{ir})]] \quad (31)$$

Thus, the initial centre of the training set can be computed as $C_1^{(0)} = (x_{i1}^{(0)}, x_{i2}^{(0)}, x_{i3}^{(0)}, x_{i4}^{(0)})$.

(ii) Initial classification $G_i^{(0)} = \{G_{i1}^{(0)}, G_{i2}^{(0)}, G_{i3}^{(0)}, G_{i4}^{(0)}\}$ is acquired through ED calculation.

(iii) New cluster centre $C_i^{(1)}$ is recalculated according to the initial grouping result obtained in step (ii)

$$C_i^{(1)} = \frac{1}{n_i} \sum G_i^{(0)} \quad (32)$$

(iv) Then get new clustering result satisfied with the condition function

$$G_i^{(1)} = \{x: d(x, c_{ij}^{(1)}) \leq d(x, c_{ik}^{(1)})\} \quad (33)$$

(v) Iterate above steps for t times until two consecutive calculations are the same, satisfying the following criteria: If $G_i^{(t)} \neq G_i^{(t-1)}$, then $t = t + 1$ and go back to step (iii); If $G_i^{(t)} = G_i^{(t-1)}$, end the iteration process and record the final grouping results.

3.3 Semi-supervised learning algorithm based on MD

Compared with ED, the correlation between variables can be described, based on Markov distance, by using covariance $\sigma_{ij} = \text{cov}(x_i, x_j)$ without being affected by dimensions. It has advantages in dealing with distance metric learning between multivariate high dimensional samples.

To make full use of useful information contained in unlabelled data, online sequential ELM-MD (OSELM-MD) is proposed in this study. By using the semi-supervised learning algorithm with the ability of reversible deduction, correctly labelled samples are applied to the extension of the new semi-labelled sample. Training

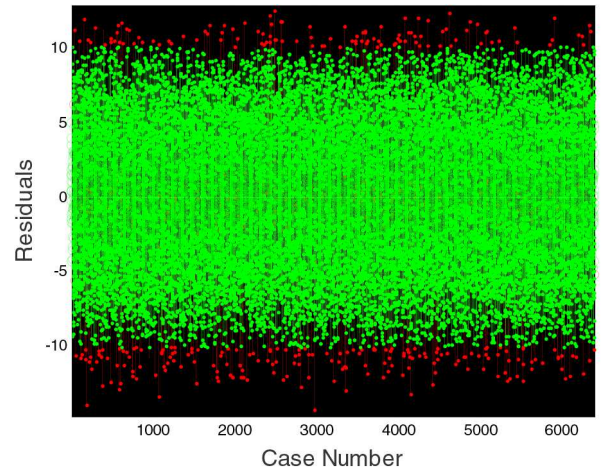


Fig. 5 Residual level diagram

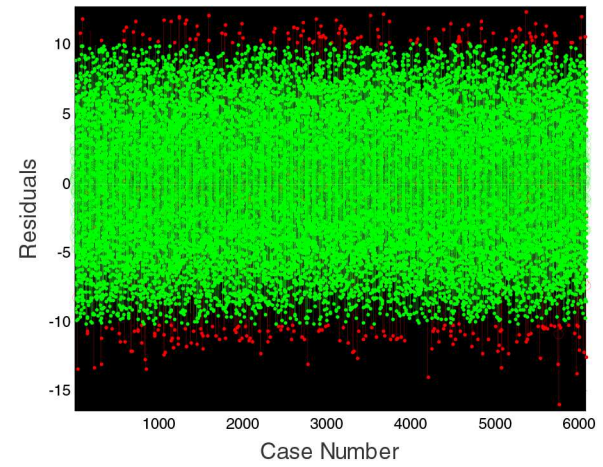


Fig. 6 Modified residual level diagram

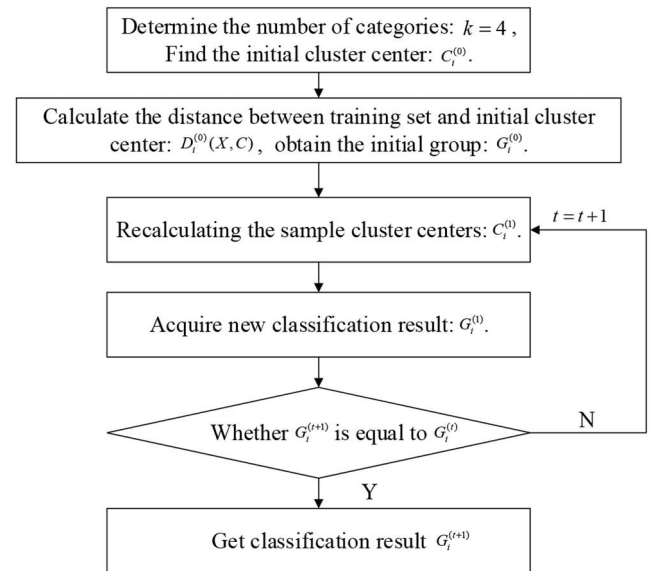


Fig. 7 Flowchart of unsupervised k-means algorithm

samples with wrong judgment are labelled as invalid samples. Then the parameters are amended.

Training set T contains tagged set T_1 and unlabelled set T_2 . The cost function of semi-supervised OSELM-MD is expressed as

$$E = \sum_{j=1}^{N_0} \left(\sum_{i=1}^{\tilde{N}} \beta_i g(\mathbf{w}_i \cdot \mathbf{x}_j^0 - b_i) - \mathbf{t}_j^0 \right)^2 + \sum_{j=1}^{N_e} \left(\sum_{i=1}^{\tilde{N}} \beta_i g(\mathbf{w}_i \cdot \mathbf{x}_j^e - b_i) - \mathbf{t}_j^e \right)^2 \quad (34)$$

where N_0 is the number of labelled samples; $\tilde{N}$ is the number of implicit strata nodes; $\mathbf{x}_j^0 = [x_{j1}^0, \dots, x_{jm}^0]^T$ is the tagged sample vector; $\mathbf{n}$ is the dimension of sample vector; $\mathbf{t}_j^0 = [t_{j1}^0, \dots, t_{jm}^0]^T$ is the class marking sample vector of $\mathbf{x}_j^0$; C is the number of output nodes of network; $\mathbf{w}_i = [w_{i1}, \dots, w_{in}]^T$ is the input weight vector between input nodes and implicit strata nodes; b_i is offset of node i ; g is the activation function of implicit strata nodes; $\beta_i = [\beta_{i1}, \beta_{i2}, \dots, \beta_{iC}]^T$ is the output weight vector between implicit strata nodes and output nodes; N_e is the number of extended samples; $\mathbf{x}_j^e = [x_{j1}^e, \dots, x_{jm}^e]^T$ is extended sample vector; $\mathbf{t}_j^e = [t_{j1}^e, \dots, t_{jm}^e]^T$ is the class marking sample vector of $\mathbf{x}_j^e$.

Get $\hat{\beta}$ that satisfies the following formula: (see (35)), where $\mathbf{H}_0$ is the hidden layer output matrix of the labelled sample set; β is the output matrix of weight; $\mathbf{T}_0$ is the class marking matrix of labelled sample; $\mathbf{H}_e$ is the hidden layer output matrix of the expanded sample; $\mathbf{T}_e$ is the class marking matrix of the expanded sample.

Pseudo-code is used to describe the process of semi-supervised machine learning as the following algorithm (Fig. 8):

The number of overall sample G index is $m=24$, and the mean vector of the high-dimensional data sample is $\mu = (\mu_1, \mu_2, \dots, \mu_{24})'$, MD between overall sample G and single sample $X = (x_1, x_2, \dots, x_{24})'$ is

$$d^2(X, G) = (X - \mu)' \Sigma^{-1} (X - \mu) \quad (36)$$

Covariance between evaluation indicator is written as

$$\Sigma = (\sigma_{ij})_{m \times m} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2m} \\ \dots & \dots & \ddots & \dots \\ \sigma_{m1} & \sigma_{m2} & \dots & \sigma_{mm} \end{bmatrix}_{m \times m} \quad (37)$$

Its solution formula is defined as

$$\begin{cases} \sigma_{ij} = \frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j) \\ \bar{x}_i = \frac{1}{n} \sum_{k=1}^n x_{ki} \\ \bar{x}_j = \frac{1}{n} \sum_{k=1}^n x_{kj} \end{cases} \quad (i, j = 1, 2, \dots, m) \quad (38)$$

3.4 Contrastive analysis

To verify the validity and accuracy of these algorithms, supervised, unsupervised, and semi-supervised machine learning methods

```

// Step 1: Initial training
(W, B, β̂₀) = Initial training(G₁, N)
// Input: labeled sample G₁, the number of hidden layer neurons Ñ
// Output: input weight matrix W, offset vector B, output weight
matrix of hidden layer β̂₀
// Random assignment for W and B
(1) for i = 1: Ñ
(2) {wᵢ, bᵢ} = rand{W, B}
(3) end
(4) Calculate output matrix of hidden layer generated by input labeled
sample:
H₀ = [ g(w₁ · x₁⁰ + b₁)  ⋯  g(w_Ñ · x₁⁰ + b_Ñ)
        ⋮                ⋱                ⋮
        g(w₁ · x_N₀⁰ + b₁)  ⋯  g(w_Ñ · x_N₀⁰ + b_Ñ) ]_{N₀ × Ñ}

(5) Calculate output weight matrix of hidden layer:
β̂₀ = K₀⁻¹ H₀ᵀ T₀ = (H_Lᵀ H_L)⁻¹ H₀ᵀ T

(6) Return W, B, β̂₀
// Step 2: Expand labeled sample

(7) Calculate output of output layer for unlabeled sample:
O_G₂ = H(W, B, G₂) β̂₀

(8) Set up the extended sample set L_e, the number of unlabeled
sample N_e.
// Step 3: Uncertainty assessment test
(9) for i = 1: N_e
(10) if 1/N₀ ||H(W, B, G₀) β̂₀ - T₀|| ≤ ρ
(11) Validation passes, turn to step 4.
(12) elseif
(13) Clear invalid sample, update the threshold
Θ = {θ_k | θ_k = θ_k + ηΔ; 0 < η < 1}
Turn to step 2
(14) end
(15) end
// Step 4: Get output calculation

(16) Calculate output weight matrix:
β̂⁽¹⁾ = [ [H(W, B, G₁)]ᵀ [H(W, B, G₁)] ]⁻¹ [H(W, B, G₁)]ᵀ T
[ [H(W, B, G₂)]ᵀ [H(W, B, G₂)] ]⁻¹ [H(W, B, G₂)]ᵀ T

(17) if ||β̂⁽¹⁾ - β̂⁽⁰⁾|| > ε && ε ≥ 0
(18) β̂⁽¹⁾ → β̂⁽⁰⁾, Turn to step 2
(19) elseif
(20) end
(21) return β̂⁽¹⁾, W, B

```

Fig. 8 Pseudo-code algorithm

$$\begin{aligned} & \| \mathbf{H}_0 \hat{\beta} - \mathbf{T}_0 \| + \| \mathbf{H}_e \hat{\beta} - \mathbf{T}_e \| = \min_{\beta} (\| \mathbf{H}_0 \beta - \mathbf{T}_0 \| + \| \mathbf{H}_e \beta - \mathbf{T}_e \|) \\ & = \min_{w_i, b_i, \beta_i} \left(\left\| \begin{bmatrix} g(w_1 \cdot x_1^0 + b_1) & \dots & g(w_{Ñ} \cdot x_1^0 + b_{Ñ}) \\ \vdots & \ddots & \vdots \\ g(w_1 \cdot x_{N_0}^0 + b_1) & \dots & g(w_{Ñ} \cdot x_{N_0}^0 + b_{Ñ}) \end{bmatrix}_{N_0 \times Ñ} \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_{Ñ}^T \end{bmatrix} - \begin{bmatrix} t_1^{0T} \\ \vdots \\ t_{N_0}^{0T} \end{bmatrix} \right\| \right. \\ & \quad \left. + \left\| \begin{bmatrix} g(w_1 \cdot x_1^e + b_1) & \dots & g(w_{Ñ} \cdot x_1^e + b_{Ñ}) \\ \vdots & \ddots & \vdots \\ g(w_1 \cdot x_{N_e}^e + b_1) & \dots & g(w_{Ñ} \cdot x_{N_e}^e + b_{Ñ}) \end{bmatrix}_{N_e \times Ñ} \begin{bmatrix} \beta_1^T \\ \vdots \\ \beta_{Ñ}^T \end{bmatrix} - \begin{bmatrix} t_1^{eT} \\ \vdots \\ t_{N_e}^{eT} \end{bmatrix} \right\| \right) \end{aligned} \quad (35)$$

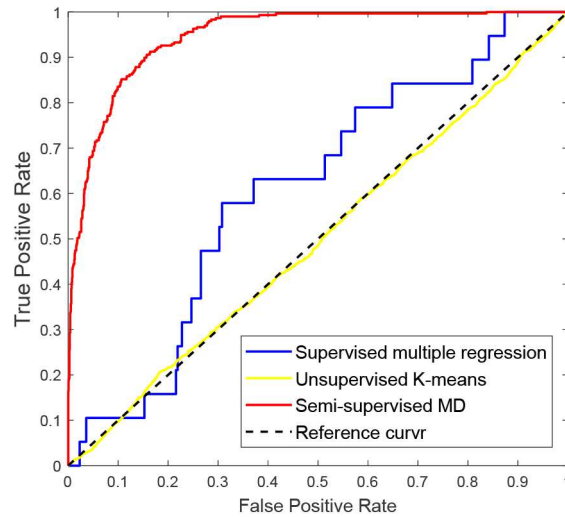


Fig. 9 ROC evaluation curve

Table 8 Algorithm comparison analysis

Device type	Algorithm	Training accuracy	Accuracy for test set, %	Process time, s	AUC
protective device	1	83.69	84.56	1.9163	0.6084
	2	63.56	61.31	2.4055	0.4952
	3	94.56	93.06	488.0919	0.9464
M-C device	1	80.50	82.31	2.0159	0.6248
	2	50.25	54.63	2.0135	0.5012
	3	97.56	96.50	496.5464	0.9584
merging unit	1	78.63	80.13	1.9246	0.6147
	2	52.19	56.44	2.9846	0.5150
	3	95.56	94.56	460.1249	0.9688
intelligent terminal	1	81.88	83.69	1.8864	0.6054
	2	61.31	59.88	2.8465	0.4984
	3	96.50	93.88	501.3124	0.9626

mentioned above are compared and analysed. To ensure the fairness of comparative analysis, a simulation test is completed on the same computer with the same sample.

Take protective devices for example, when the scale of the training set is $s = 80\%$, the receiver operating characteristic (ROC) evaluation curve results of three algorithms are shown in Fig. 9.

As shown in Fig. 9, the black dotted line of three graphs is the reference curve of random guess which is considered as the reference curve classified with 50% probability. The closer the ROC curve is the top left corner of the graph, the closer the area under the curve (AUC) value gets to 1, which means the corresponding machine learning method has higher prediction performance. The AUC value of supervised multiple regression analysis algorithm is $0.6084 \in (0.5, 1)$, then the prediction result is better than random prediction with certain reference significance. The AUC value of unsupervised k -means algorithm based on ED is $0.4952 \rightarrow 0.5$, i.e. to say, the prediction result is similar to random prediction with meaningless reference significance. The AUC value of semi-supervised learning algorithm based on MD is $0.9464 \rightarrow 1$, i.e. the prediction result of the algorithm is excellent.

The algorithm comparison analysis table is presented in Table 8.

In regard to the accuracy, processing time, and feasibility, the calculation result of three machine learning algorithms for four kinds of protective devices is shown in Table 8. Supervised multiple regression analysis algorithm takes 1.8864–2.0159 s to get the result. Unsupervised k -means algorithm based on ED takes 2.0135–2.9846 s to gain the result. A semi-supervised learning algorithm based on Markov distance takes 460.1249–501.3124 s to attain the result. Compared with semi-supervised learning algorithm based on Markov distance, supervised multiple regression analysis algorithm, and unsupervised k -means algorithm based on ED spend less time on getting calculation result, but their

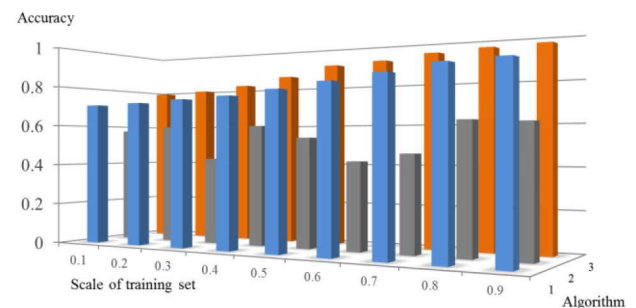


Fig. 10 Machine learning algorithm comparison analysis

accuracy is far less than that of semi-supervised learning algorithm based on Markov distance. At the same time, the AUC value is around 0.95, which is obviously better than of the other two methods.

The scale of the training set is one of the main factors influencing the precision of the machine learning algorithm. Nine training subsets whose scale is from 10 to 90% are selected in the same sample set. The precision analysis is shown in Fig. 10.

The influence of the training scale on accuracy is shown in Fig. 10. As for Algorithms 1 and 3, i.e., supervised multiple regression analysis algorithm and semi-supervised learning algorithm based on Markov distance, the accuracy is sensitive to the scale of training set when the sample is insufficient. $\Delta r = 1 - r_i/r_{\max}$ ($i = 10, 20, \dots, 90\%$) represents the sensitive threshold of the algorithm. When the scale of the training set is over 70%, evaluation accuracy is no longer sensitive to the scale for algorithm 1 ($\Delta r_1 < 5\%$). When the scale of the training set is over 60%, evaluation accuracy is no longer sensitive to the scale for Algorithm 3 ($\Delta r_3 < 5\%$). However, accuracy is not subject to the

training scale for Algorithm 2. It can be seen from the picture that Algorithm 3 has the lowest requirement for the scale of the training sets. When the scale reaches 60%, it can meet the requirement to get excellent evaluation precision.

4 Maintenance strategy

4.1 Maintenance strategy

Maintenance strategy of power system relay protection can be divided into four levels in terms of severity and urgency:

- **Level I:** The integrity of relay protection equipment, namely the entire screen and whole device, should be inspected, repaired, replaced, and tested, with new equipment being verified within one year; power-cut maintenance.
- **Level II:** Partial domain inspection, maintenance, replacement, and experiment for relay protection equipment; defect checking and handling should be carried out by taking advantage of primary devices' blackout opportunity; power-cut maintenance.
- **Level III:** Routine inspection, maintenance, and test for relay protection equipment should be carried out; power-cut maintenance.
- **Level IV:** Non-power cut inspection and maintenance for relay protection equipment.

4.2 Determination of maintenance strategy

Different maintenance strategies will be adopted in light of the corresponding evaluation results:

- **Normal state:** Arrange level III maintenance during primary equipment blackout. The reference cycle can be postponed for a year depending on the actual operation condition of equipment; level IV maintenance can be arranged appropriately before level III maintenance.
- **Warning state:** Arrange level II maintenance during primary equipment blackout or arrange level III maintenance depending on the actual operation condition of equipment; level IV maintenance and surveillance should be strengthened before outage maintenance.
- **Error state:** Arrange level I or level II maintenance appropriately in view of equipment defect type and elimination time; if caused by the deterioration of multiple different modules, level I maintenance should be arranged. If caused by the deterioration of an individual module, level II maintenance should be arranged; level IV maintenance and surveillance should be strengthened before outage maintenance.
- **Critical state:** Corresponding emergency measures should be adopted immediately and level I or level II maintenance should be scheduled at once; if caused by the deterioration of multiple different modules, level I maintenance should be arranged. If caused by the deterioration of an individual module, level II maintenance should be arranged; level IV maintenance and surveillance should be strengthened before outage maintenance.

5 Conclusion

This study developed a risk assessment model for relay protection systems based on a semi-supervised MD machine learning algorithm. First, to provide training sets for machine learning method, a state evaluation model on the basis of AHP-fuzzy synthetic theory is performed on M-C device, merging unit, protective equipment, and intelligent terminal. Then, to make the evaluation results more objective and accurate, machine learning algorithms including supervised multiple regression analysis, unsupervised k -means, and semi-supervised MD method are applied and compared in the state evaluation for relay protection systems. Finally, the simulation results indicate that the semi-supervised MD machine learning algorithm has higher precision and lower requirements for the scale of data set.

Future work will further consider the prediction for the potential loss of power systems caused by the failure of the relay protection system and its relevant preventive measures.

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